Skills Summary

Evaluating Perfect Square Roots and Cube Roots

Use this reference while completing your worksheet or when studying.

A root is a question about equal factors. $\sqrt{n} = s$ means $s \cdot s = n$; $\sqrt[3]{n} = s$ means $s \cdot s \cdot s = n$. The index says *how many equal factors*.

Squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225

Cubes: 1, 8, 27, 64, 125, 216, 343, 512, 729, 1000

If the radicand is in the matching list the root is exact; if it is not, the root is irrational and can only be bracketed and rounded.

1. Perfect square roots

Ask "what times itself?" and read the square list. For a radicand past the list, split it into perfect-square factors and root each piece: $\sqrt{ab} = \sqrt{a}\sqrt{b}$, e.g. $\sqrt{400} = \sqrt{4}\sqrt{100} = 20$.

Example: Evaluate $\sqrt{196}$.
Step 1. The question asks for a factor used twice, so scan the square list instead of dividing — division is the operation that produces the classic wrong answer here.
Step 2. $14 \cdot 14 = 196$, and 196 is in the list, so the root is exact.

$$\sqrt{196} = \mathbf{14}$$

Try it: Evaluate $\sqrt{121}$.

check: **11**

2. Perfect cube roots

Ask "what times itself *three* times?" and read the cube list. Cubes grow fast, so the list is short — $\sqrt[3]{1000} = 10$ is the last one worth memorising. Unlike square roots, cube roots also work on negatives: $\sqrt[3]{-8} = -2$.

Example: Evaluate $\sqrt[3]{64}$.
Step 1. The index is 3, so the cube list is the right table — 64 appears in both lists, which is exactly why the index has to be read first.
Step 2. $4 \cdot 4 \cdot 4 = 64$, so the cube root is 4 (while the *square* root of 64 is 8).

$$\sqrt[3]{64} = \mathbf{4}$$

Try it: Evaluate $\sqrt[3]{125}$.

check: **5**

3. Roots of fractions and decimals

A root distributes over a quotient: $\sqrt{\dfrac{a}{b}} = \dfrac{\sqrt{a}}{\sqrt{b}}$ and $\sqrt[3]{\dfrac{a}{b}} = \dfrac{\sqrt[3]{a}}{\sqrt[3]{b}}$.

For a decimal, rewrite it as a fraction first: $0.49 = \frac{49}{100}$, $0.008 = \frac{8}{1000}$. Whatever you do on top you must do underneath.

Example: Evaluate $\sqrt{\dfrac{9}{16}}$.

Step 1. Both parts are in the square list, so splitting the root across the fraction turns one hard root into two easy ones.

Step 2. $\dfrac{\sqrt{9}}{\sqrt{16}} = \dfrac{3}{4}$, and squaring back gives $\frac{9}{16}$.

$$\mathbf{\tfrac{3}{4}}$$

Try it: Evaluate $\sqrt{0.49}$ exactly.

check: **0.7**

4. Estimating a root that is not perfect

Trap the radicand between the two nearest entries of the matching list, then the root sits between their roots:

$$p^2 < n < q^2 \implies p < \sqrt{n} < q$$

Refine by squaring a guess. A rounded decimal is a *stand-in*: it is rational, the root is not, so write $\approx$, never $=$.

Example: Estimate $\sqrt{30}$ to two decimal places.

Step 1. 30 is not in the square list, so bracket it first rather than hunting for an exact value that does not exist.

Step 2. $25 < 30 < 36$ gives $5 < \sqrt{30} < 6$; testing $5.5^2 = 30.25$ shows the root is just under 5.5, and to two places it is 5.48.

$$\sqrt{30} \approx \mathbf{5.48}$$

Try it: Estimate $\sqrt{90}$ to two decimal places.

check: **9.49**

(!) Watch out: taking a root is not dividing. $\sqrt{144} \neq 72$ and $\sqrt[3]{27} \neq 9$ — check by multiplying your answer back out. And read the index before the radicand: 64 is in both lists, so $\sqrt{64} = 8$ while $\sqrt[3]{64} = 4$.